

OAK RIDGE National Laboratory | MANUFACTURING DEMONSTRATION FACILITY

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Adamantine Overview and Tutorials

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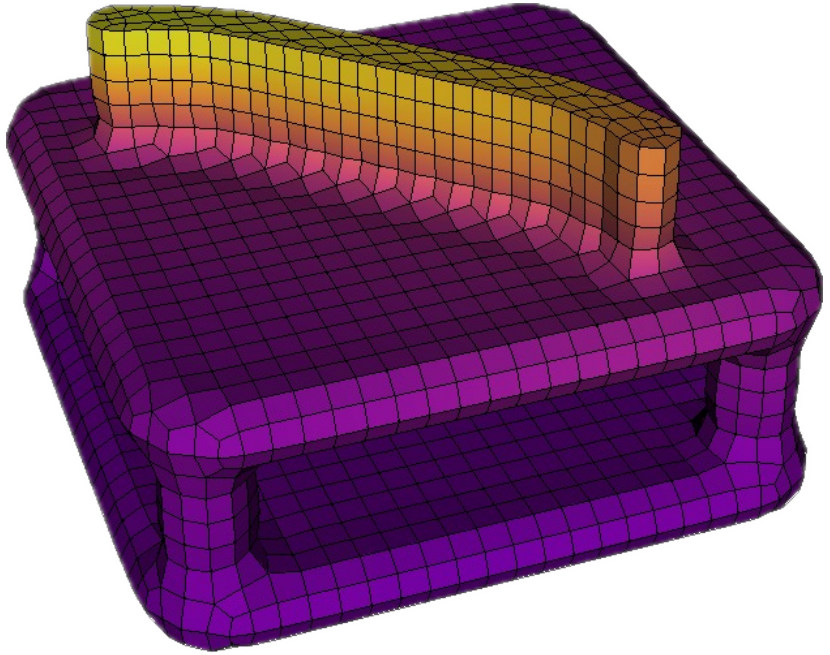
Computational Science and Engineering

U.S. DEPARTMENT of ENERGY

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What is Adamantine?

- Finite element thermomechanical code for AM
- Use temperature-dependent material properties with phase change
- Explicit in time
- Use adaptive mesh refinement
- Integrates with SCOPS
- GPU support
- License: Apache 2.0 with LLVM exception (LLVM license)
- GitHub: https://github.com/adamantine-sim/adamantine
- Website: https://adamantine-sim.github.io/adamantine



Our goal: Monitoring printing with a real-time digital twin

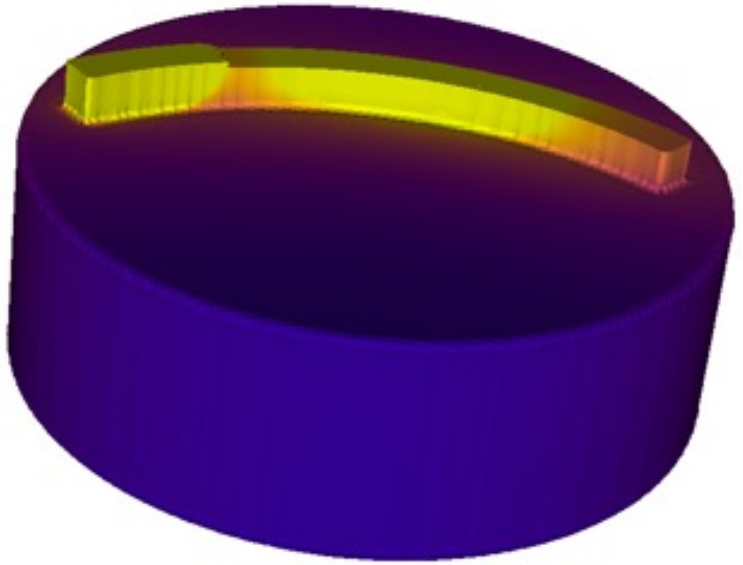
In-situ observations



Tied to reality
Only part of the outer surface

+

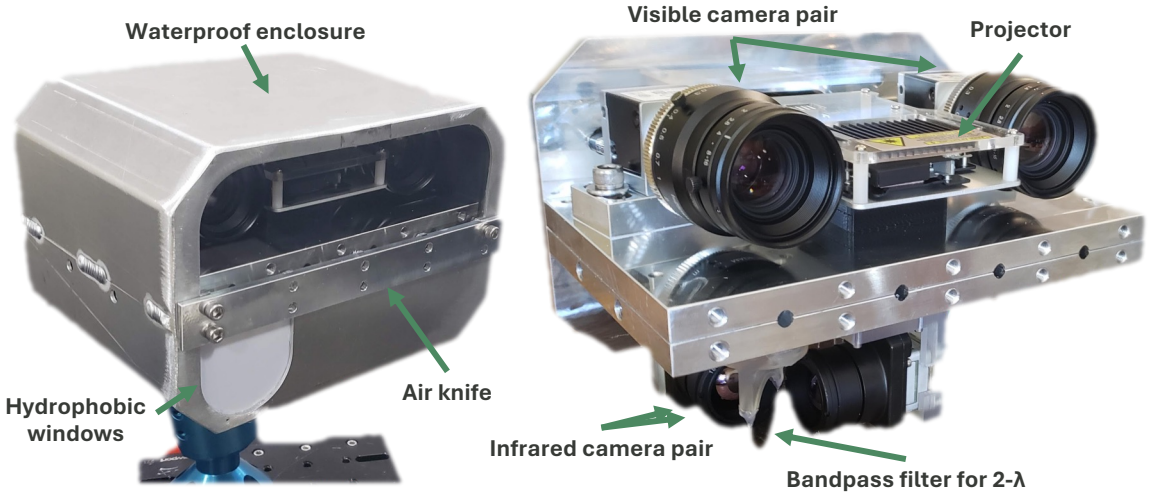
Simulations

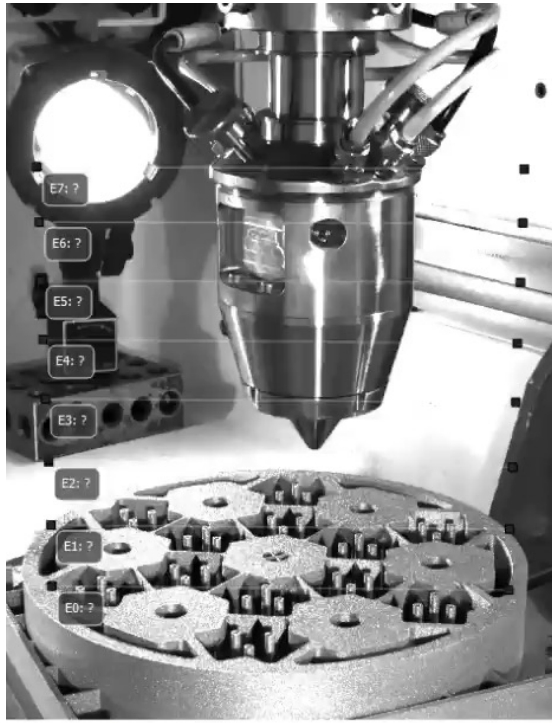


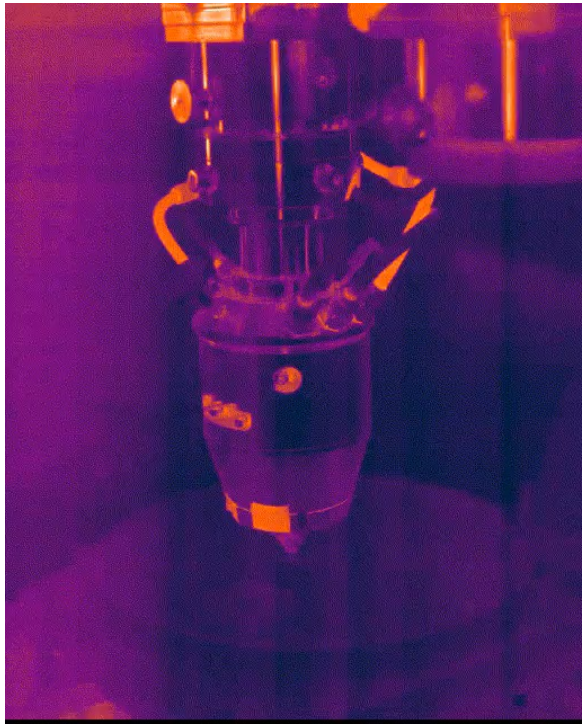
Not tied to reality
Full 3D volume

Together, estimate tied to reality for the full 3D volume

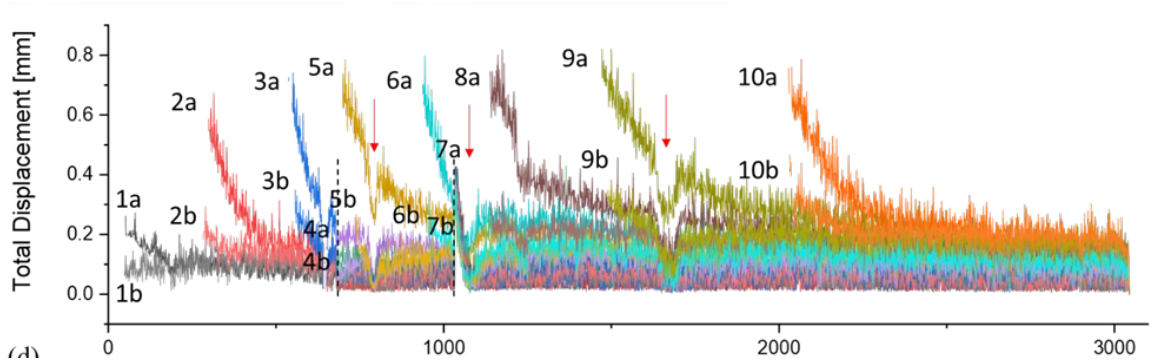
SCOPS: Stereo Correlated Optical and Pyrometric Sensing





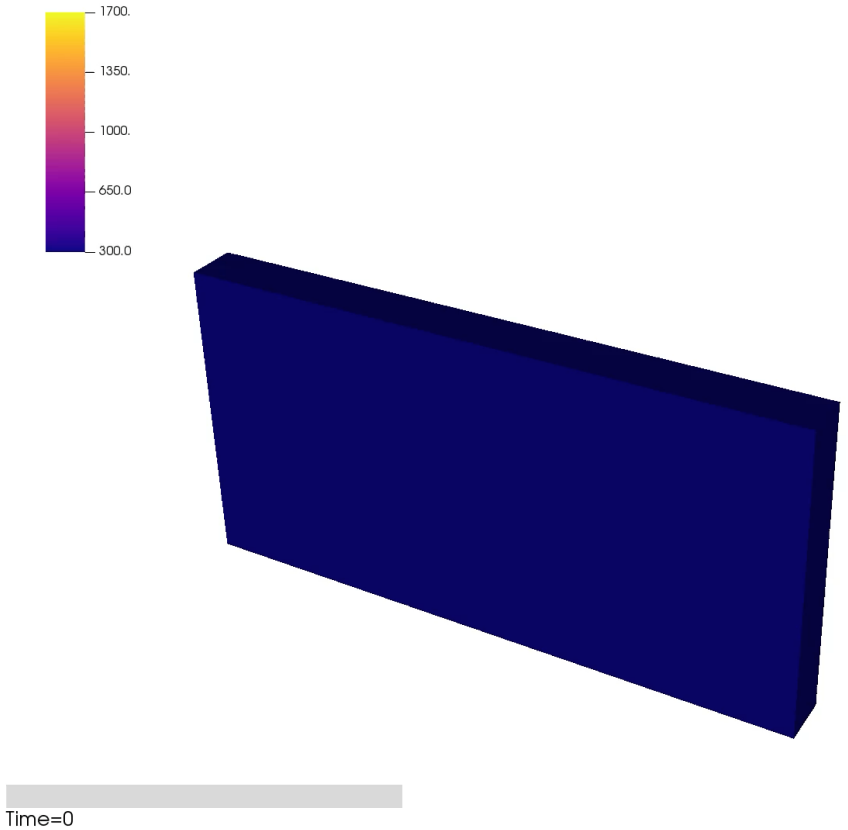


- Directed Energy Deposition welding induces distortion.
  Control requires measurement.
  - DIC tracks surface roughness features in 3D, IR tracks temperature over time
  - *Thermoplastic* history observed
- Imaging enhances simulation.
  - Validation, or *assimilation*
- Works for all DED classes



Faster than real time simulation

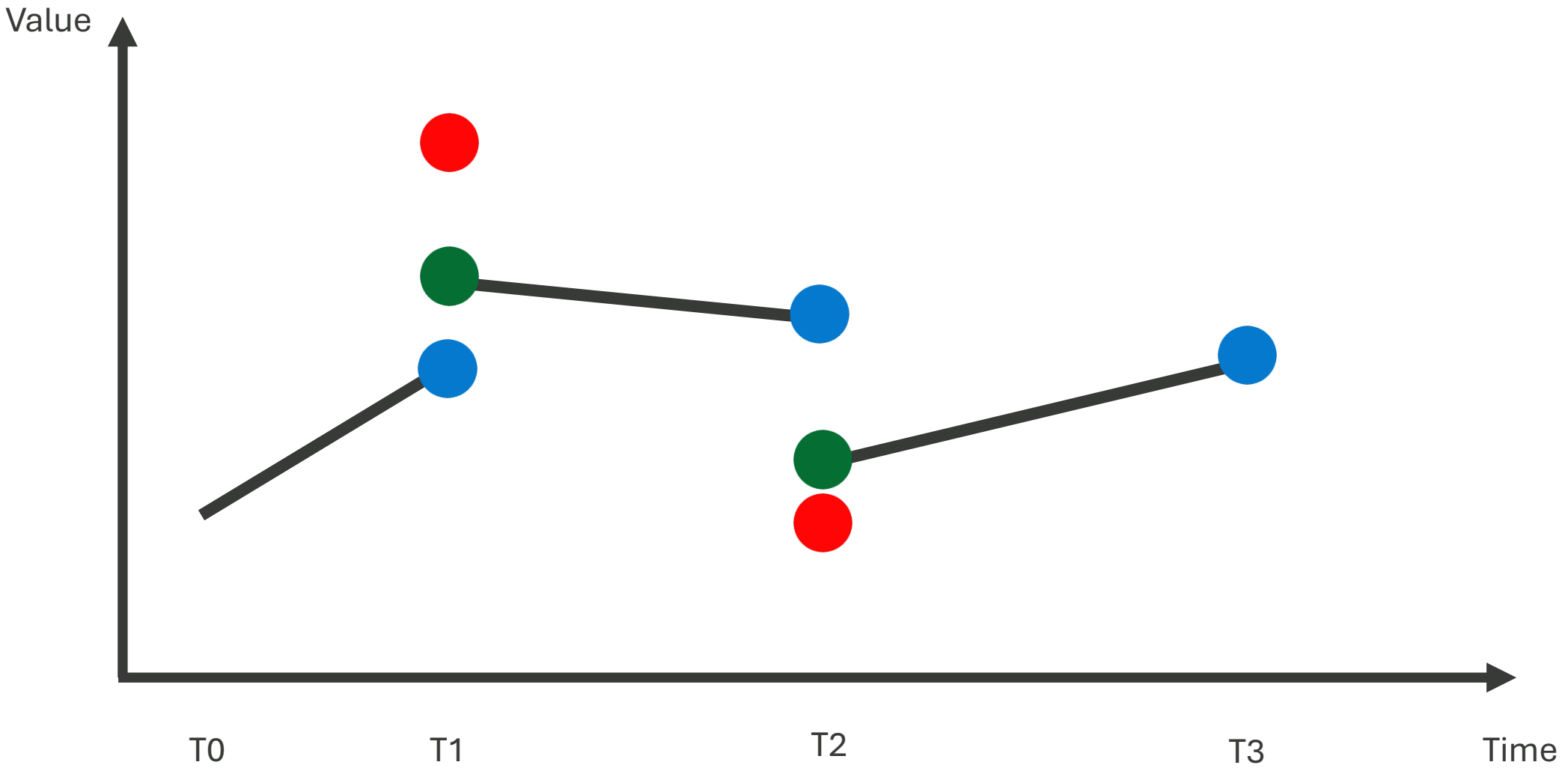
- Fast simulations are necessary to
  - Assimilate data in real-time
  - Close the loop and have simulations guide the printing process
- Simulation must be at least 3X faster than real time
- Extensive use of matrix-free algorithms and vectorization



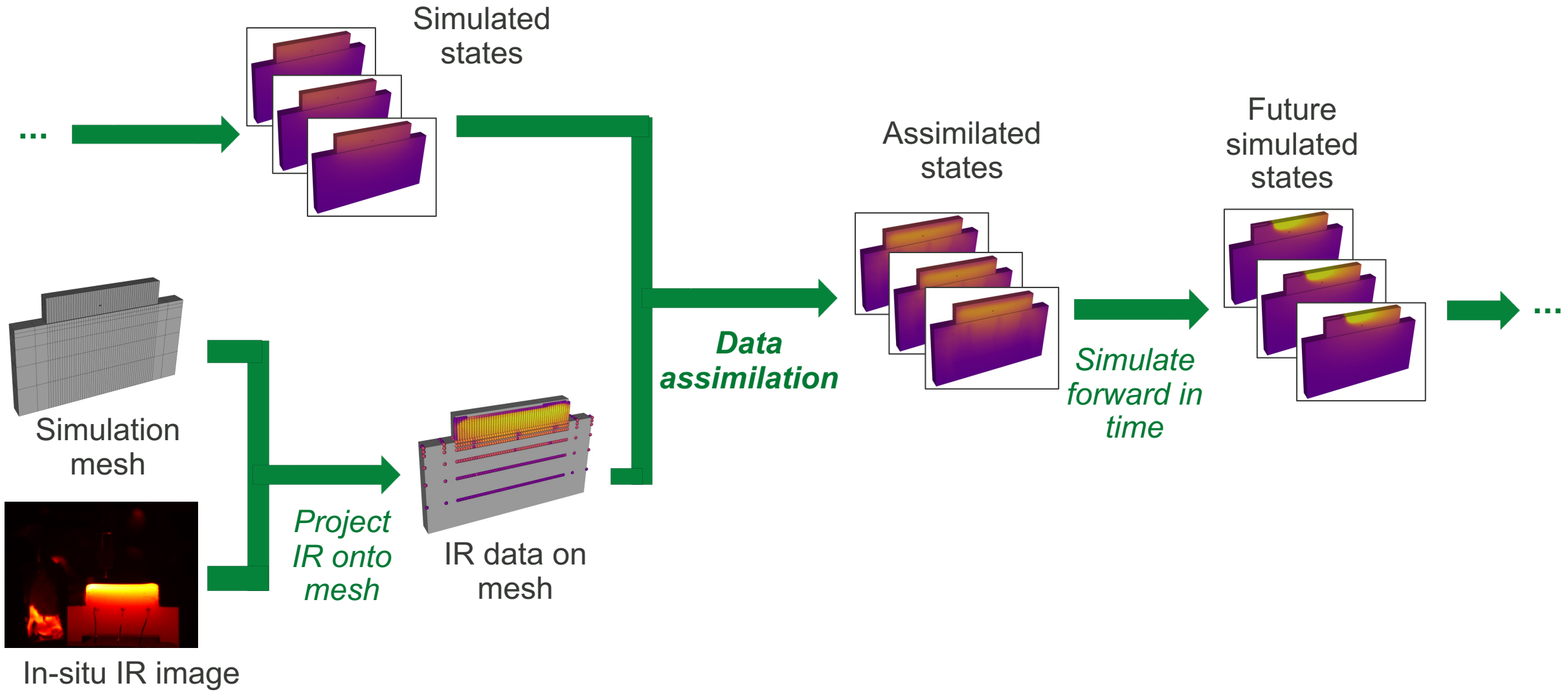
Combining experimental data and simulation with data assimilation

- “Data assimilation is the approximation of the true state of some physical system at a given time by combining time-distributed observations with a dynamical model in an optimal way” (Data Assimilation Methods, Algorithms, and Applications by Marc Asch, Marc Bocquet, and Maelle Nodet)
- First developed for weather forecast
- Two main classes of methods: statistical methods and variational methods

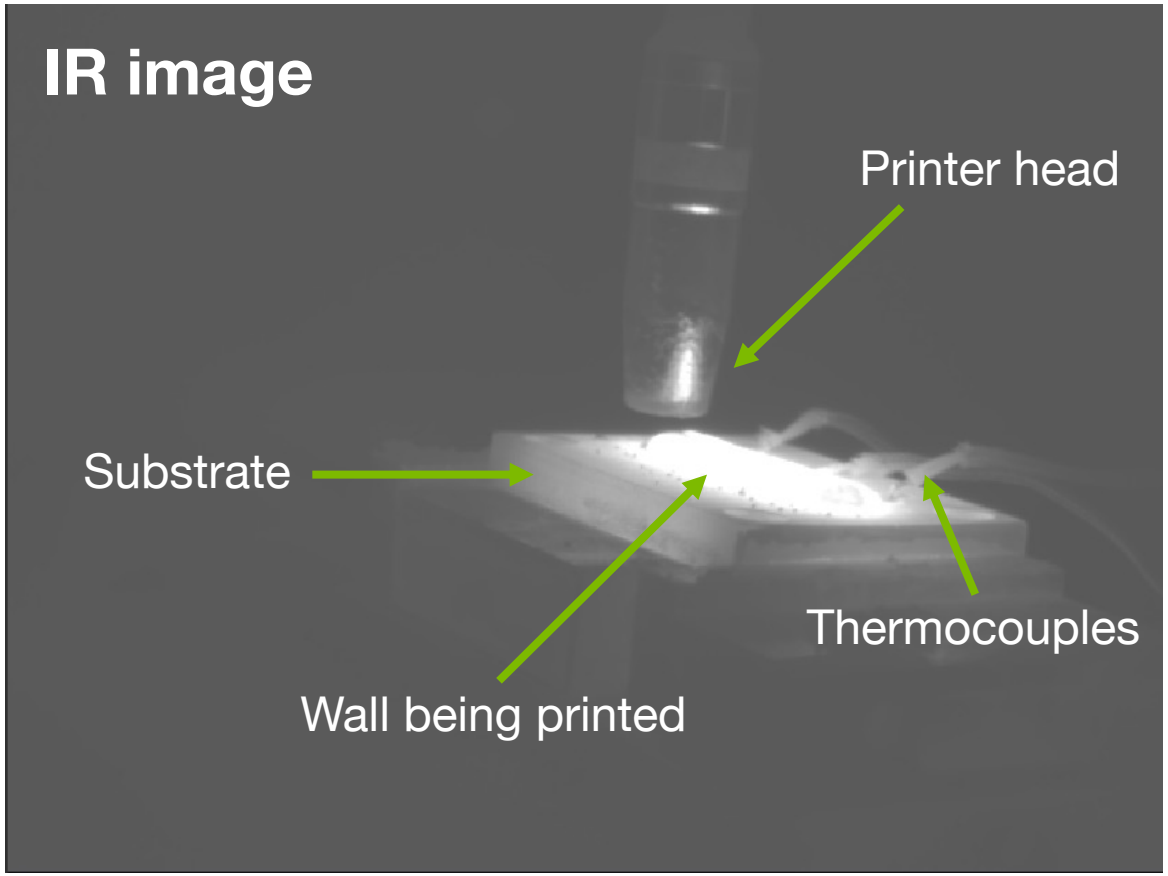
Combining experimental data and simulation with data assimilation

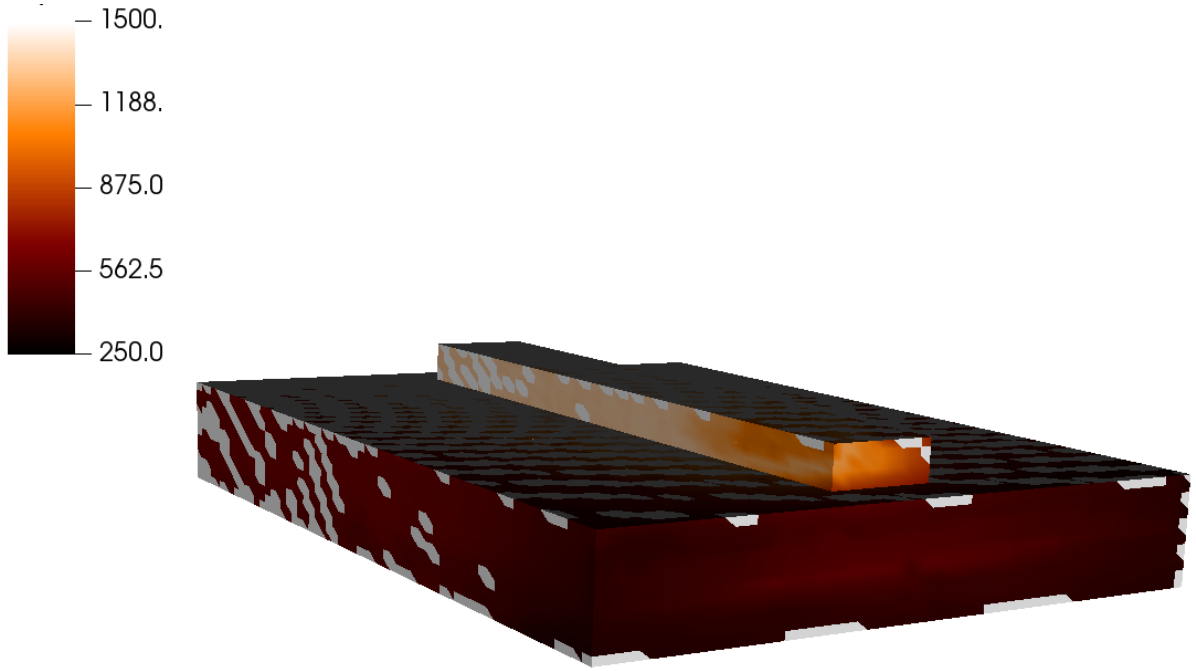


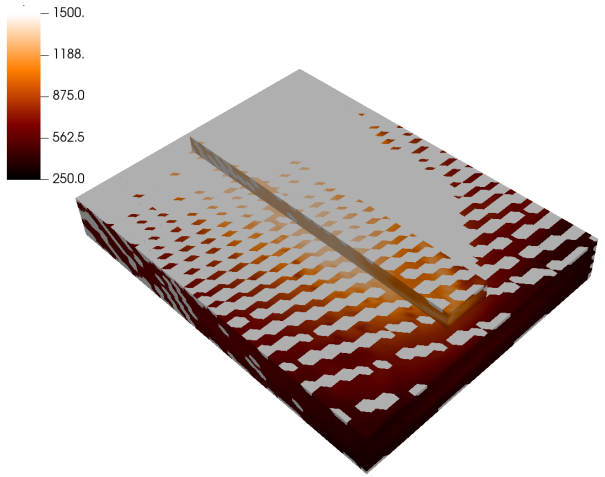
Data assimilation workflow



Projecting the IR data onto the simulation mesh





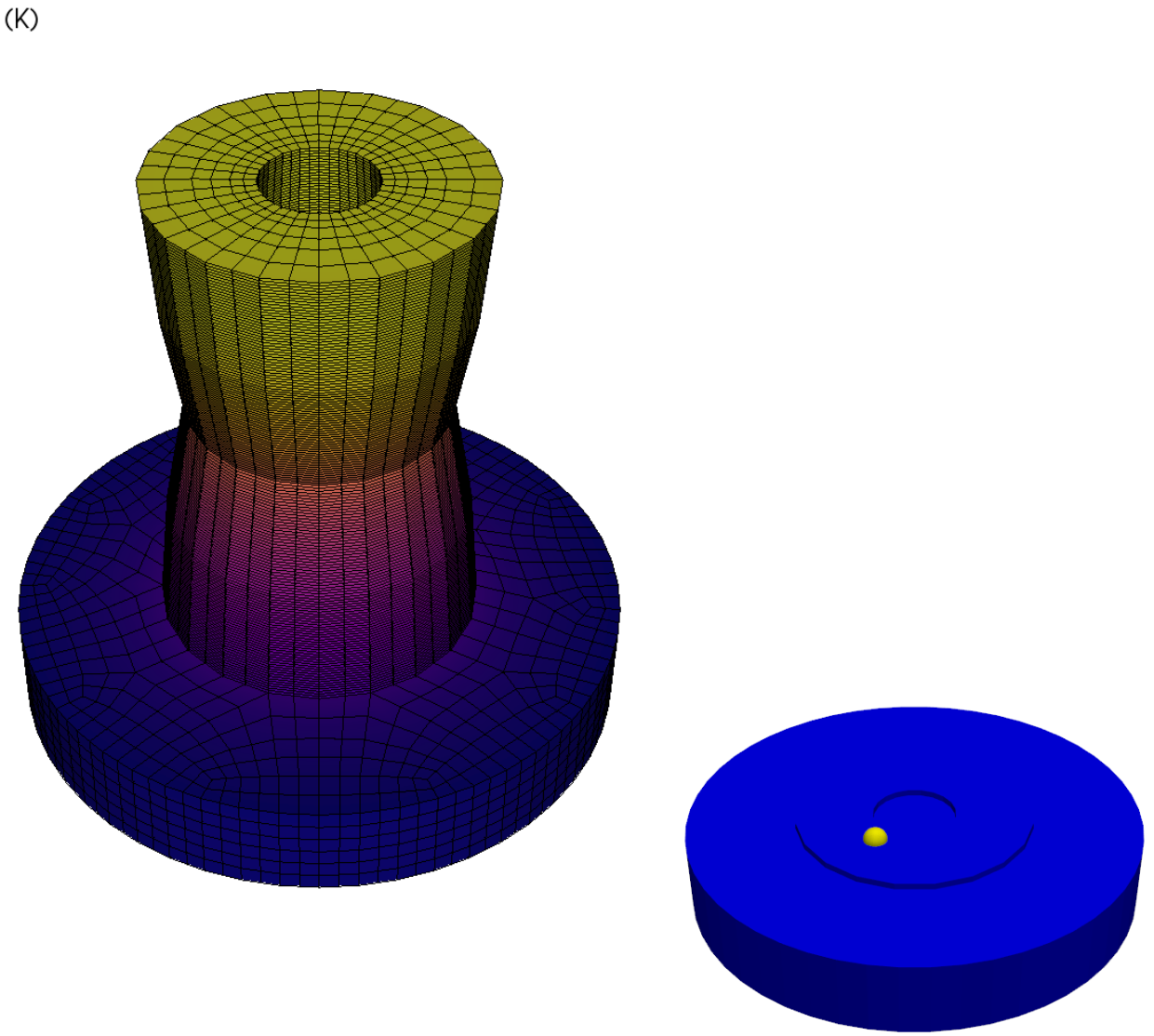


Coordinate transformations nominally handled in ROS

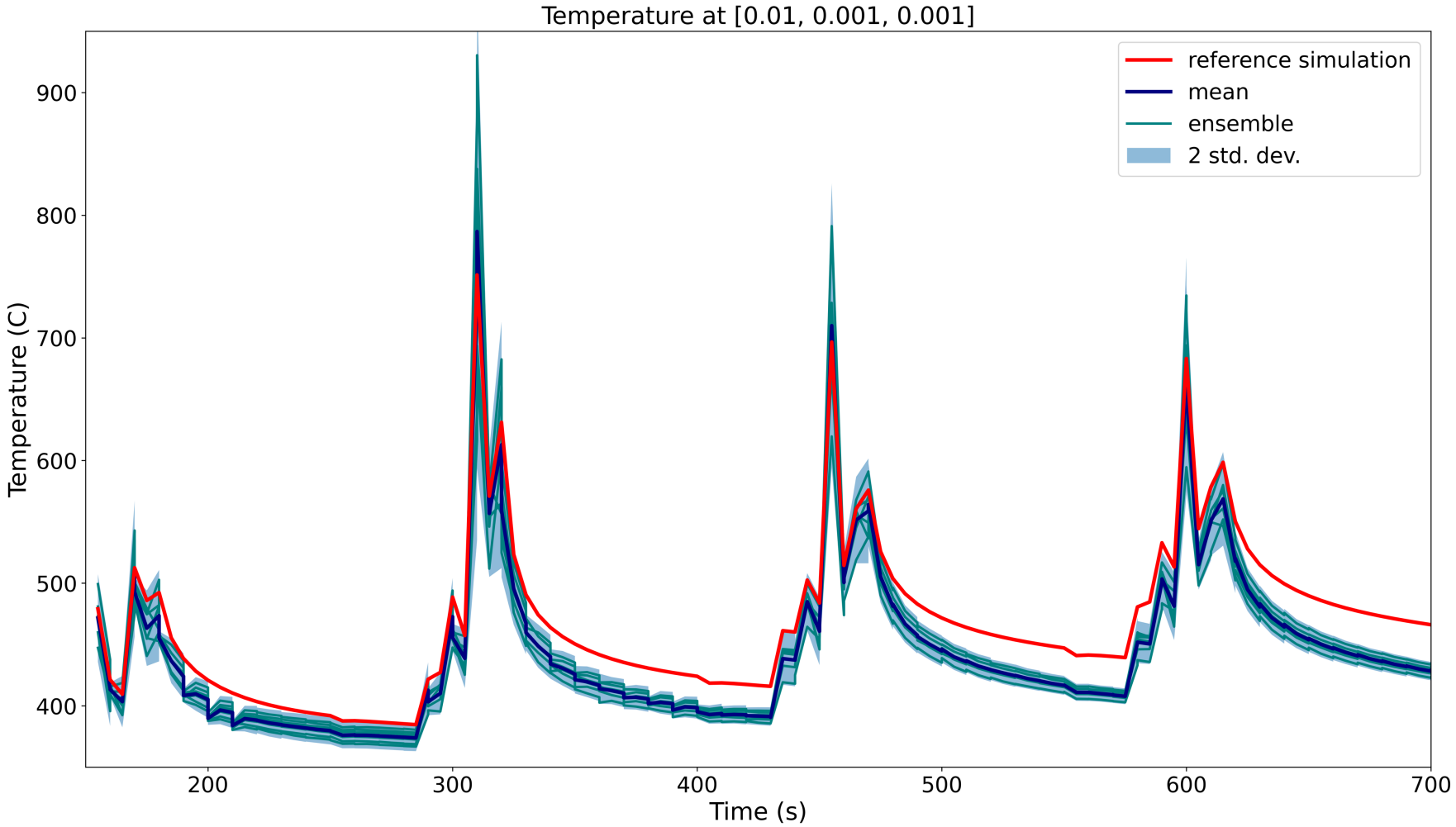
Camera coordinates -> Substrate coordinates

Data assimilation: real test case

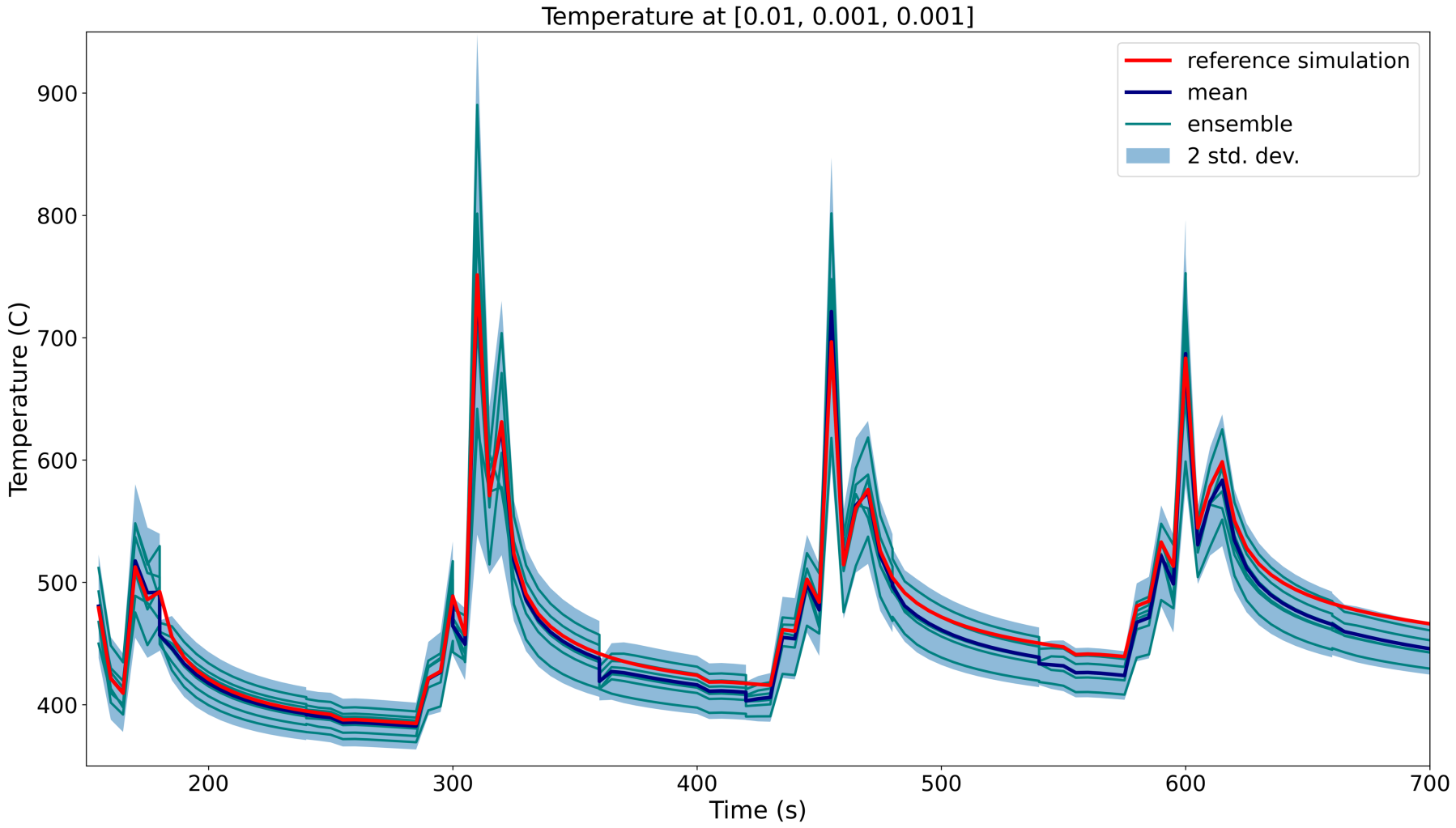
- Experiment done September 2025
- Material: stainless steel
- Experimental time: ~3h
- Simulation time: ~1h



Data assimilation every 10 seconds



Data assimilation every 60 seconds

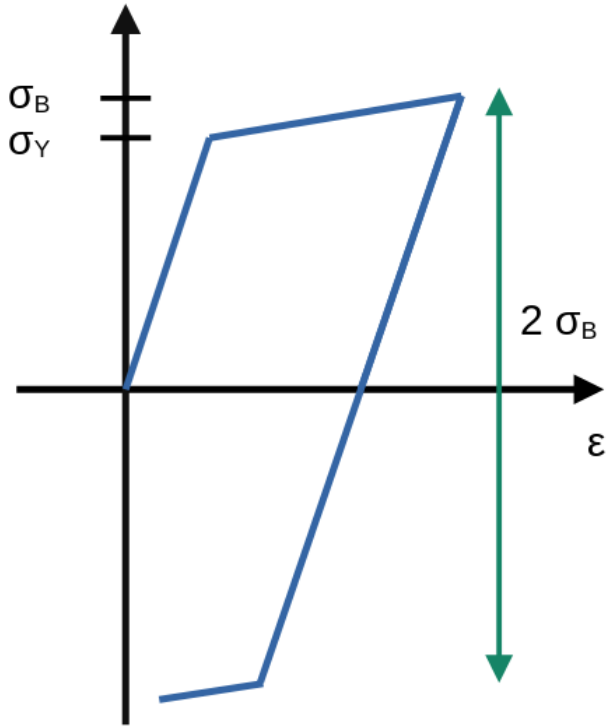


Thermomechanical simulation

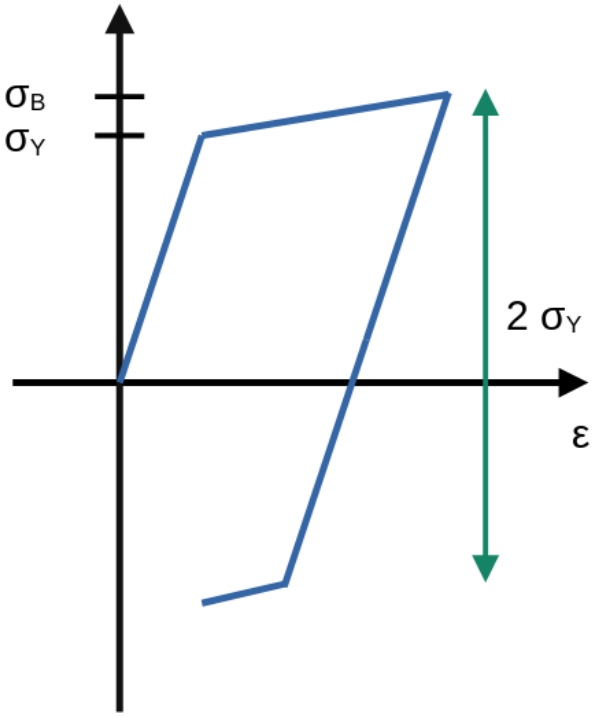
- Material properties must be isotropic → this is not true for thermal simulation
- Infinitesimal displacement → we do not support large deformations
- One-way coupling from the temperature evolution to the mechanical evolution → neglect the effect of deformation on the thermal simulation
- Plasticity constitutive laws supported: isotropic and kinematic hardening
- Currently comparing results with Abaqus

Plasticity constitutive laws

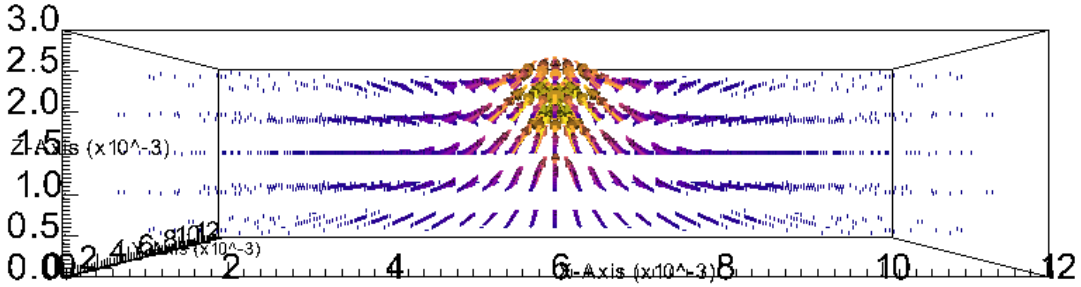
Isotropic hardening: uniform expansion of the material's yield surface in stress space

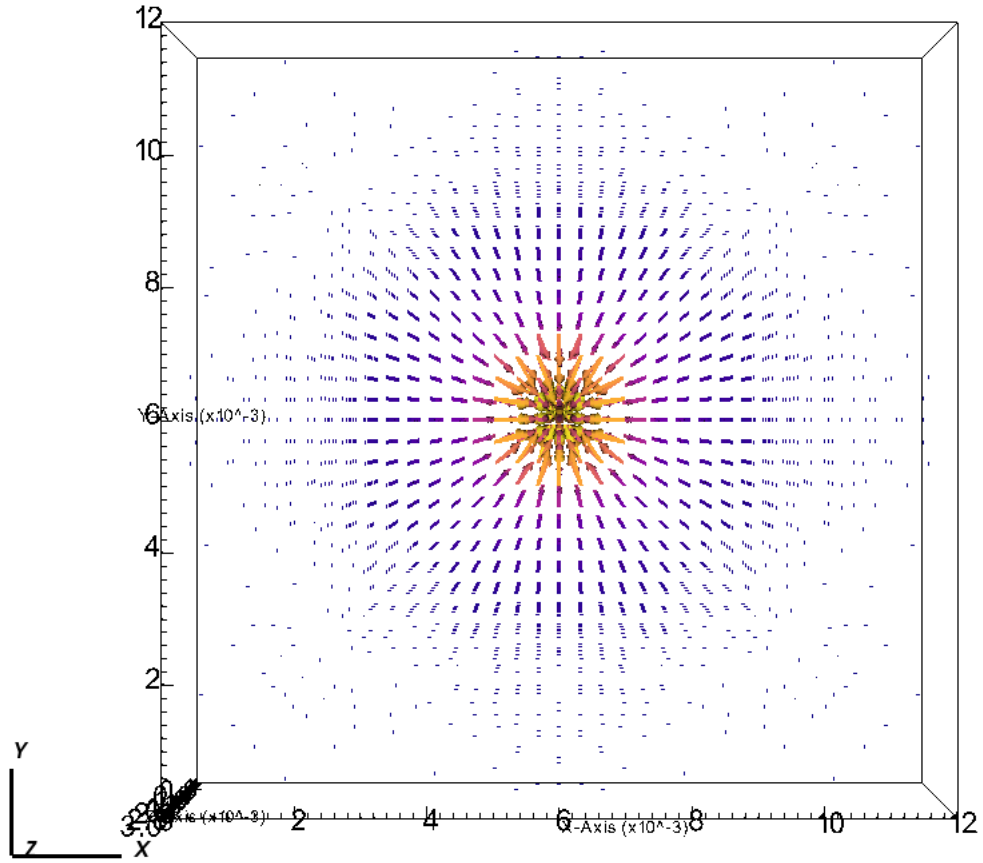


Kinematic hardening: yield surface in stress space moves without any change in its shape or size



Thermoelastic simulation: displacement field





How to use Adamantine

Visualization

- Output files produced by a simulation:
  - **.vtu:** output data for a given time step and a given processor
  - **.pvtu:** one file per time step to assemble the .vtu from different processors
  - **.pvd:** one file per simulation to assemble the different .pvtu files
- Inactive cells have a temperature of 0K
- To only visualize the active cells:
  - In VisIt: use Threshold operator with a minimal temperature
  - In ParaView: use the Threshold filter with a minimal temperature

Input file

- Based on Boost Property Tree: arbitrarily nested tree of keys-values
- Supported formats:
    - info: Boost format, supports comments
    - json: file extension must be .json
- By default, use SI units but this can be changed in the input file

Input file

```
geometry
{
  import_mesh false ; Use built-in mesh generator
  dim 2 ; dimension of the domain
  length 2e-2 ; [m]
  height 1e-2 ; [m] In 3D, the third parameters is width
  length_divisions 20 ; Number of cell layers in the length direction
  height_divisions 10 ; Number of cell layers in the height direction
}

physics
{
  thermal    true  ; Thermal simulation
  mechanical false ; Mechanical simulation. If both thermal and mechanical are
                   ; true, solve a coupled thermo-mechanical problem
}

boundary
{
  type adiabatic
}
```

Required inputs

- **geometry:** simulation domain
- **boundary:** boundary conditions
- **time_stepping:** time step size and end time of the simulation
- **sources:** heat sources
- **materials:** material properties
- **discretization:** order the finite element used
- **physics:** thermal and/or mechanical simulation
- **post_processor:** filename and output frequencies
- **refinement:** number of refinement cycles and frequency between refinement

Optional inputs

- **ensemble:** parameters to create an ensemble of simulations with input parameters that follow a normal distribution
- **checkpoint:** filename and frequency of checkpoints
- **data_assimilation:** parameters used by EnKF
- **experiment:** information about the experimental data
- **memory_space:** decide between CPU and GPU simulation
- **microstructure:** filename for temperature gradient and cooling rate
- **profiling:** caliper configuration
- **restart:** restart filename
- **units:** choose non-default units of input entries
- **verbose_output:** turn on/off the verbosity

Exercise 1: Geometry

- Describe the mesh used by the simulation
- Mesh can be 2D or 3D
- Box mesh can be generated from input file
- Load a mesh from mesh generator: hexahedra only, no hanging nodes
- Many formats supported: abaqus, gmsh, tecplot, exodus, etc.
- Need at least one active cells at the start of the simulation
- Describe the material deposition: length, width, height, time, etc.

Exercise 1: Geometry

- Exercise is in the ***work/02_Adamantine/01*** directory
- The instructions are in the README.md file
- Use https://adamantine-sim.github.io/adamantine/doc/input_file.html#geometry-required
- Run adamantine using: *adamantine –i input.info*
- The solution of the exercise is given in the solution.info file
- You can also try to load the gmsh file: *domain.msh*

Exercise 2: Boundary and time stepping

- Boundary:
  - Thermal boundary condition supported are: adiabatic, radiative, and convective
  - A boundary can be both radiative and convective
  - A general boundary can be applied to the entire domain boundary
  - One can overwrite the boundary condition for any boundary ids
- Time stepping:
  - Set the time step size
  - Set the end time of the simulation or until the scan path ends

Exercise 2: Boundary and time stepping

- Exercise is in the ***work/02_Adamantine/02*** directory
- The instructions are in the README.md file
- Use https://adamantine-sim.github.io/adamantine/doc/input_file.html#boundary-required and https://adamantine-sim.github.io/adamantine/doc/input_file.html#time_stepping-required
- Run adamantine using: *adamantine –i input.info*
- The solution of the exercise is given in the solution.info file

Exercise 3: Heat Source

- Adamantine supports an arbitrary number of heat sources but the heat sources must be in the same z-plane
- Two types of sources are supported: goldak and electron beam
- Sources must be enumerated consecutively starting at zero
- Two scan path formats are supported: segment format and event format
- Segment format is similar to the format used by AdditiveFoam but the power coefficient multiplies the maximum power set in the input file

Exercise 3: Heat source

- Exercise is in the ***work/02_Adamantine/03*** directory
- The instructions are in the README.md file
- Use https://adamantine-sim.github.io/adamantine/doc/input_file.html#sources-required
- Run adamantine using: *adamantine –i input.info*
- The solution is given in the solution.info file

Exercise 4: Material Properties

- Materials properties can be set for powder, solid, and liquid state
- Adamantine solves the anisotropic heat equation → the thermal conductivity needs to be given for x, y, and z (follows deposition axis)
- It is not necessary to set all the material properties
- Material properties can be given as a polynomial of the temperature or as a table
- Important:
    - Define a material for each material id in the mesh
    - Material ids are consecutive numbers starting at 0

Exercise 4: Material Properties

- Exercise is in the ***work/02_Adamantine/04*** directory
- The instructions are in the README.md file
- Use https://adamantine-sim.github.io/adamantine/doc/input_file.html#materials-required
- Run adamantine using: *adamantine –i input.info*
- The solution is given in the solution.info file